

Agentic RAG over SEC 10-K Filings

Sample Prompts — Evidence of Agent Behaviour

Screenshots of the three demo prompts running end to end in the Streamlit interface. Each page shows the prompt, the agent's answer with citations, and the expanded reasoning trace.

1. Multi-Hop Planning

The planner splits one question into two independently filtered sub-questions, retrieves each separately, and the decide node confirms both companies' facts are present before synthesising a single grounded comparison.

2. The Clarify Gate

No company or year was given. The agent queries the DuckDB catalog, finds the request matches all ten filings, and stops to ask rather than guessing.

3. Answering After Clarification

The clarification is appended to the original question, so the agent answers what was actually asked. Every figure is cited back to a specific chunk, filing and Item section.

4. Dynamic Action Selection

The decide node chooses 'refine', rewrites its own search query and retries with a wider net. When the second attempt still lacks the fact, it chooses 'answer' and reports honestly that the filings do not contain it, rather than inventing one.

Prompt

“Compare Apple and Tesla risk factors in 2024”

The planner splits one question into two independently filtered sub-questions, retrieves each separately, and the decide node confirms both companies' facts are present before synthesising a single grounded comparison.

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Ingested Corpus

10 filings indexed

AMAZON COM INC (AMZN) FY2024

AMAZON COM INC (AMZN) FY2025

Apple Inc. (AAPL) FY2024

Apple Inc. (AAPL) FY2025

MICROSOFT CORP (MSFT) FY2024

MICROSOFT CORP (MSFT) FY2025

NVIDIA CORP (NVDA) FY2024

NVIDIA CORP (NVDA) FY2025

Tesla, Inc. (TSLA) FY2024

Tesla, Inc. (TSLA) FY2025

Suggested Prompts

Compare Apple and Tesla risk factors in 2024

What were the main revenue drivers?

What was Microsoft's CEO's exact favorite color in 2024?

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I currently have access to the following filings:

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Here are a few things you can ask me:

- Compare Apple and Tesla risk factors in 2024 (Demonstrates Multi-Hop Planning)
- What were the main revenue drivers? (Demonstrates the Clarify Gate)
- What was Microsoft's CEO's exact favorite color in 2024? (Demonstrates Context Verification)

Compare Apple and Tesla risk factors in 2024

To compare Apple and Tesla risk factors in 2024, we can look at the passages provided.

Apple's risk factors include:

- Design and manufacturing defects that could materially adversely affect the Company's business and result in harm to the Company's reputation [chunk 110, Apple Inc. FY2024, Item 1A].
- Investment in new business strategies and acquisitions could disrupt the Company's ongoing business, present risks not originally contemplated and materially adversely affect the Company's business, reputation, results of operations and financial condition [chunk 120, Apple Inc. FY2024, Item 1A].
- The Company's retail stores are subject to numerous risks and uncertainties [chunk 120, Apple Inc. FY2024, Item 1A].
- Legal and regulatory compliance risks, including the risk of unfavorable results of legal proceedings or government investigations [chunk 129, Apple Inc. FY2024, Item 1A].
- Risks associated with changes in the value of the U.S. dollar relative to local currencies [chunk 129, Apple Inc. FY2024, Item 1A].
- Credit risk and fluctuations in the values of its investment portfolio [chunk 125, Apple Inc. FY2024, Item 1A].

Tesla's risk factors include:

- Delays in launching and ramping the production of products and features, or inability to control manufacturing costs [chunk 1139, Tesla, Inc. FY2024, Item 1A].
- Concentration of risk, including credit risk, supply risk, and inventory valuation risk [chunk 1238, Tesla, Inc. FY2024, Item 3].
- Dependence on suppliers, including single source suppliers, and the inability of these suppliers to deliver necessary components of products in a timely manner [chunk 1238, Tesla, Inc. FY2024, Item 3].

Both companies face risks related to their products and services, but Apple's risks seem to be more focused on design and manufacturing defects, investment risks, and regulatory compliance, while Tesla's risks are more focused on production delays, supply chain risks, and concentration of risk.

However, it's worth noting that these passages do not provide a comprehensive list of risk factors for either company, and there may be other risks that are not mentioned here. [chunk 1139, Tesla, Inc. FY2024, Item 1A] and [chunk 110, Apple Inc. FY2024, Item 1A] both mention that the risks described are not the only risks facing the company.

Reasoning

Plan: 2 sub-question(s)

- "AAPL risk factors 2024" -> AAPL / 2024
- "TSLA risk factors 2024" -> TSLA / 2024

Clarify: All targets resolved – searching

Retrieved 4 chunks for "AAPL risk factors 2024" (Attempt 1)

Retrieved 4 chunks for "TSLA risk factors 2024" (Attempt 1)

Decide: answer – The passages contain the raw facts needed to compare Apple and Tesla risk factors in 2024.

Answer: Synthesized response

Prompt

“What were the main revenue drivers?”

No company or year was given. The agent queries the DuckDB catalog, finds the request matches all ten filings, and stops to ask rather than guessing.

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What were the main revenue drivers?

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- What were the main revenue drivers? (Demonstrates the Clarify Gate)
- What was Microsoft's CEO's exact favorite color in 2024? (Demonstrates Context Verification)

What were the main revenue drivers?

Which filing did you mean for "main revenue drivers"? Options: Apple Inc. (AAPL) FY2024, AMAZON COM INC (AMZN) FY2025, Tesla, Inc. (TSLA) FY2025, MICROSOFT CORP (MSFT) FY2025, NVIDIA CORP (NVDA) FY2025, Apple Inc. (AAPL) FY2025, AMAZON COM INC (AMZN) FY2024, Tesla, Inc. (TSLA) FY2024, MICROSOFT CORP (MSFT) FY2024, NVIDIA CORP (NVDA) FY2024

Reasoning

Plan: 1 sub-question(s)
1. "main revenue drivers" -> any / any year
Clarify: 10 matches - asking user

Prompt

“Nvidia 2025”

The clarification is appended to the original question, so the agent answers what was actually asked. Every figure is cited back to a specific chunk, filing and Item section.

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- Compare Apple and Tesla risk factors in 2024 (Demonstrates Multi-Hop Planning)
- What were the main revenue drivers? (Demonstrates the Clarify Gate)
- What was Microsoft's CEO's exact favorite color in 2024? (Demonstrates Context Verification)

What were the main revenue drivers?

Which filing did you mean for "main revenue drivers"? Options: Apple Inc. (AAPL) FY2024, AMAZON COM INC (AMZN) FY2025, Tesla, Inc. (TSLA) FY2025, MICROSOFT CORP (MSFT) FY2025, NVIDIA CORP (NVDA) FY2025, Apple Inc. (AAPL) FY2025, AMAZON COM INC (AMZN) FY2024, Tesla, Inc. (TSLA) FY2024, MICROSOFT CORP (MSFT) FY2024, NVIDIA CORP (NVDA) FY2024

Reasoning

Nvidia 2025

The main revenue drivers for Nvidia in 2025 were [chunk 845, NVIDIA CORP FY2025, Item 7]:

- Compute & Networking revenue, which increased by 159% due to strong demand for accelerated computing and AI solutions, particularly from Data Center computing.
- Data Center revenue, which grew 162% driven by demand for the Hopper computing platform used for large language models, recommendation engines, and generative AI applications.
- Graphics revenue, although it decreased by 13%, was driven by sales of the GeForce RTX 40 Series GPUs.

These revenue drivers contributed to Nvidia's total revenue of \$130.5 billion in fiscal year 2025, which was up 114% from the previous year [chunk 839, NVIDIA CORP FY2025, Item 7].

Reasoning

Plan: 5 sub-question(s)
1. "AMZN FY2025 revenue drivers" -> AMZN / 2025
2. "AAPL FY2025 revenue drivers" -> AAPL / 2025
3. "MSFT FY2025 revenue drivers" -> MSFT / 2025
4. "NVDA FY2025 revenue drivers" -> NVDA / 2025
5. "TSLA FY2025 revenue drivers" -> TSLA / 2025
Clarify: All targets resolved - searching
Retrieved 4 chunks for "AMZN FY2025 revenue drivers" (Attempt 1)
Retrieved 4 chunks for "AAPL FY2025 revenue drivers" (Attempt 1)
Retrieved 4 chunks for "MSFT FY2025 revenue drivers" (Attempt 1)
Retrieved 4 chunks for "NVDA FY2025 revenue drivers" (Attempt 1)
Retrieved 4 chunks for "TSLA FY2025 revenue drivers" (Attempt 1)
Decide: answer - The passages contain the raw facts needed to answer the question about Nvidia's 2025 revenue drivers.
Answer: Synthesized response

Prompt

“What was Microsoft’s CEO’s exact favorite color in 2024?”

The decide node chooses 'refine', rewrites its own search query and retries with a wider net. When the second attempt still lacks the fact, it chooses 'answer' and reports honestly that the filings do not contain it, rather than inventing one.

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Here are a few things you can ask me:

- Compare Apple and Tesla risk factors in 2024 (Demonstrates Multi-Hop Planning)
- What were the main revenue drivers? (Demonstrates the Clarify Gate)
- What was Microsoft's CEO's exact favorite color in 2024? (Demonstrates Context Verification)

What was Microsoft's CEO's exact favorite color in 2024?

I couldn't find the answer to your question in the provided passages. The passages do not mention the CEO's favorite color in 2024 or any other year. They provide information about Microsoft's financial performance, business segments, and corporate social responsibility, but do not include personal details about the CEO's preferences. [chunk 345, MICROSOFT CORP FY2024, Item 1] does mention the CEO, Satya Nadella, but only provides information about his role, age, and position with the company.

Reasoning

Plan: 1 sub-question(s)
1. "Microsoft CEO favorite color 2024" -> MSFT / 2024
Clarify: All targets resolved - searching
Retrieved 4 chunks for "Microsoft CEO favorite color 2024" (Attempt 1)
Decide: refine - The passages contain information about Microsoft's CEO, but not their favorite color.
retry: "Microsoft CEO favorite color 2024 (MSFT / 2024) using official company terminology" -> MSFT / 2024
Retrieved 8 chunks for "Microsoft CEO favorite color 2024 (MSFT / 2024) using official company terminology" (Attempt 2)
Decide: answer - The passages contain information about Microsoft's CEO, Satya Nadella, but do not mention his favorite color.
Answer: Synthesized response